Guilherme Parreira da Silva

Data Scientist // Statistician

guilhermeparreira.silva@gmail.com +5541984352569 Curitiba, PR, Brazil

EXPERIENCE

Senior Data Scientist

4KST

July 2024 – Today

- Developed a generic Machine Learning (ML) pipeline for time series analysis that can fit any time series. It was tested in Porto Seguro (Improving 63 of 73 product lines); ExxonMobil to predict fuel consumption, where we achieved better metric results than AWS Canvas Forecasting; and Volvo to predict machine line production, improving the current metric by 15%.
- Credit score for the Lojas Colombo retail store, improving the current model by 5 percentage points in KS, doubling approved clients, and halving default rates.
- Developed a fraud risk score to support the onboarding decisions for new cards and checking accounts for the Itau bank. For cards, the bad rate decreased from .39% to.07%, increasing the consumer approval by 4 percentage points. For checking accounts, the bad rate decreased from .31% to .22%, while expanding approval by 2 percentage points.
- Developed Anti-fraud score for Sem Parar, where we achieved an impressive 92 points in KS, allowing us to cut off 50% of fraud, rejecting .03% of transactions, saving BRL 15K/month.

Data Scientist

Lightera (Furukawa Electric LatAm)

Feb 2023 - July 2024

- Streamlined an ongoing demand planning project, reducing computation time by 50% through optimized multi-core code and another 50% by changing the AWS SageMaker instance for an insignificant cost. Reduce the error by 10% by tweaking the pipeline parameters after a deep exploratory data analysis.
- Designed a machine learning model to predict the likelihood of clients and non-clients purchasing in the upcoming month. Achieving an impressive 85% AUC metric, contributing to sales revenue growth, and increasing 20 percentage points in the percentage of opportunities closed.
- Developed a predictive model to forecast cash flow, facilitating proactive expense management and scenario planning. Achieved a 90% accuracy for 1-month forecasts and over 75% accuracy for 12-month forecasts.

Data Scientist

Marista Group

April 2021 – Feb 2023

- Developed a time series model to forecast student numbers across grades and schools, distinguishing between new and existing students throughout the enrollment campaign. This pioneering data science initiative was presented via Power BI alongside an interactive simulator, enabling marketing and sales teams to optimize their advertisement budgets effectively.
- Designed a dynamic target curve for enrollment based on historical data, using a logistic model. It provided daily targets on a smoother curve for each grade, replacing the previous weekly targets with abrupt changes that did not have information by grade.

AWARDS

Data science hackathon from IBM - 2017
Member of winner team which developed a platform to manage health data to PR state

Regional Statistical Council
Best student of the class in statistics in 2016

Integrated week of teaching, research, and extension - UFPR - 2015
Best work presented about the Laboratory of Applied Statistics

Dean's Congratulation Award
Award granted for the great grades in the first semester in UWA - 2014

SKILLS

- Python
- R
- Time Series
- Ubuntu
- Supervised learning
- Not supervised learning
- GIT
- SQL
- Power BI
- Bash
- AWS

LANGUAGES

- Portuguese (Native)
- English (Fluent)

- Built a sentiment prediction model for comments from students' parents on NPS surveys with an impressive 80% recall and precision. Additionally, the model identified the topics of discussion with 80% precision across five key topics. This automation speeds up data analysis from a three-month manual process to immediate insights, allowing for a comprehensive review of all comments.

Statistical consultant

GP Solutions in Statistics

April 2017 – Oct 2021

- Statistical consultancy impacting over 200 researchers and different companies.
- Problem understanding, statistical modeling, storytelling elaboration, and presenting results.
- Statistical analysis: Multivariate analysis, Regression models, Survey data and sample size calculation, Non-parametric tests, hypothesis tests, A/B tests, time series analysis.
- As the sole employee I was responsible for all the steps, from client hunting up to maintenance services.

Junior Credit Analyst

HSBC

Dec 2015 – April 2017 // Mar – Dec 2015: Intern

- Modeling of credit risk portfolio.
- Time series study to predict revenues.
- Client segmentation using business rules.

EDUCATION

Deep learning specialization

DeepLearning.AI-Coursera

April 2026

Master in numerical methods

Federal University of Paraná (UFPR)

Jan 2019 – July 2021

Implementation of a multivariate regression model for count data.

Bachelor in Statistics

UFPR

Jan 2010 – Dec 2015

- Volunteer at the Applied Statistics Laboratory (LEA) project, which provides statistical consulting/advising for UFPR graduate students and other researchers. For each analysis, it was necessary to understand the researcher's demands and solve them using different statistical techniques, such as regression analysis and clustering for various fields of knowledge (Psychology, Zootechnics, Pharmacy, and Medicine, among others).
- My undergraduate thesis was about control charts for the mean via *Neoteric Ranked Set Sampling*.

University of Western Australia (UWA)

Jul 2013 – Dec 2014 Perth, Australia

Exchange program I studied English, programming, and statistical units.